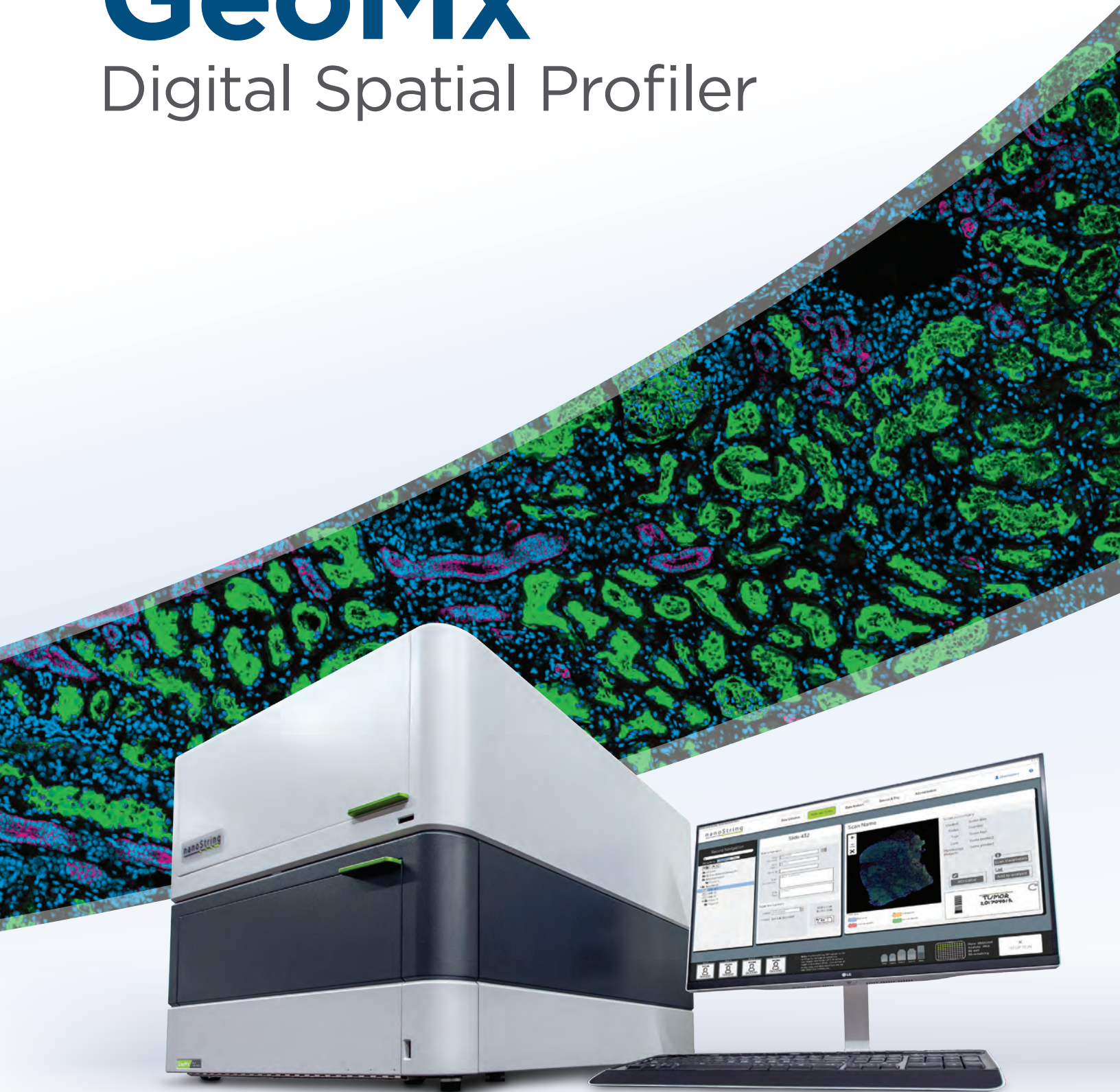


# GeoMx<sup>®</sup>

## Digital Spatial Profiler



[nanosttring.com](https://nanosttring.com)

FOR RESEARCH USE ONLY. Not for use in diagnostic procedures.

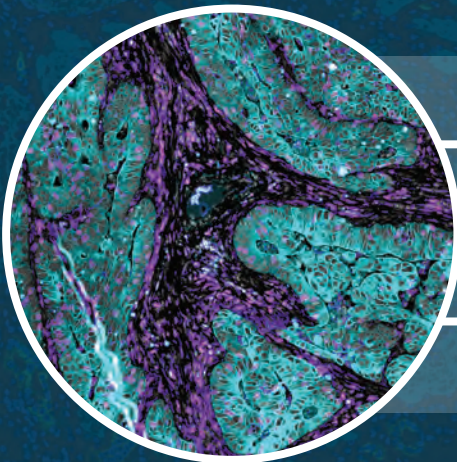
nanosttring  
A BRUKER COMPANY

# Heterogeneity Resolved with Specificity

## Spatially resolve tissue cell populations with functional segmentation

Understanding tissue heterogeneity is crucial to studying developmental biology, disease pathogenesis, and response to treatment. Bridging the gap between tissue imaging and molecular profiling technologies such as single cell analysis, the GeoMx® Digital Spatial Profiler (DSP) allows you to unlock novel biological insights with spatial multiomics.

### *In situ* Visualization Technology (FISH/ IHC)



### Molecular Profiling Technologies



|   |              |   |
|---|--------------|---|
| + | Spatial      | - |
| - | Plex         | + |
| - | Quantitation | + |
| - | Precision    | + |

Introducing

**GeoMx<sup>®</sup>**

# Digital Spatial Profiler

GeoMx DSP seamlessly enables spatial transcriptomics and proteomics with a workflow that combines standard pathology and molecular profiling with efficient data analysis.

GeoMx RNA assays enable quantitative, spatial analysis of 100s of transcripts up to the whole transcriptome from a single section of FFPE or fresh frozen tissue.

GeoMx Protein assays enable targeted spatial profiling of 100s of proteins from FFPE tissue sections.

GeoMx DSP also enables Spatial Proteogenomics by combining both Protein and RNA assays on the same slide.



**HETEROGENEITY,  
RESOLVED**

**UNLOCK YOUR  
SAMPLES WITH  
CONFIDENCE**

**CONSISTENT  
RESULTS,  
RELIABLE  
ANSWERS**

**DETECT MORE  
WITHOUT  
COMPROMISE**

**ANALYZE  
TODAY, PUBLISH  
TOMORROW**

**STRUCTURE  
DICTATES  
FUNCTION**

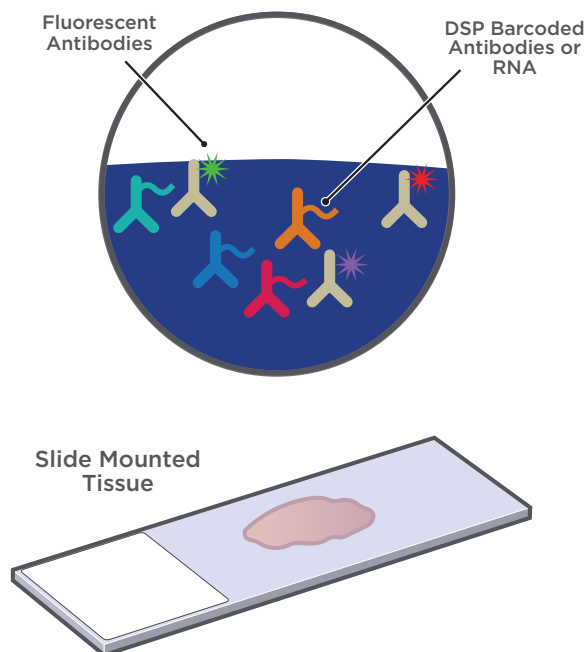
# The Path is Clear

## GeoMx<sup>®</sup> DSP Workflow

Using standard IHC methodologies, tissue sections are stained with a mixture of fluorescently-labeled and DNA-barcoded antibodies. Once imaging and profiling is complete, GeoMx DSP stores data from each region of interest (ROI) after the expression of a target is quantified.

### 1 Sample Prep\*

### 2 Profile



**Any sample, FFPE or fresh frozen, use any morphology marker, detect RNA and/or Protein.**

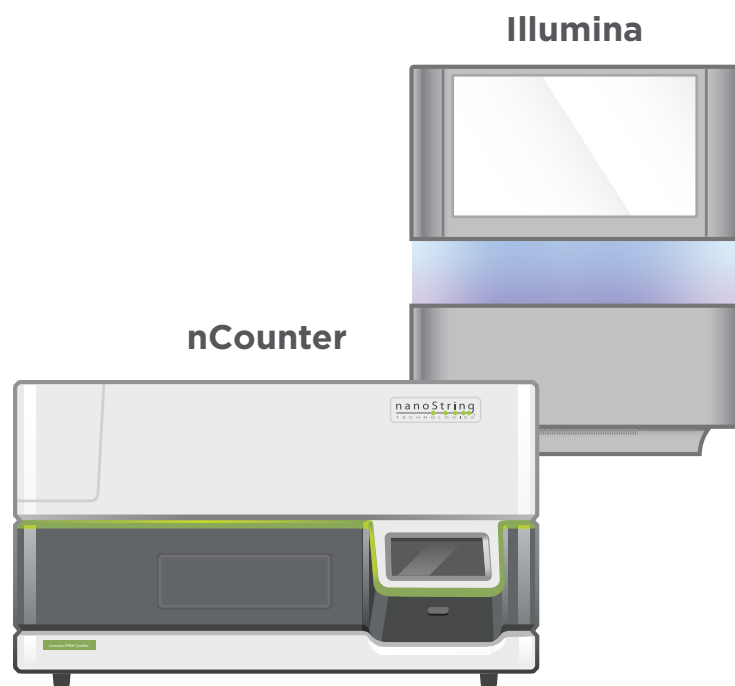
\*Can be automated.

## GeoMx<sup>®</sup> Digital Spatial Profiler



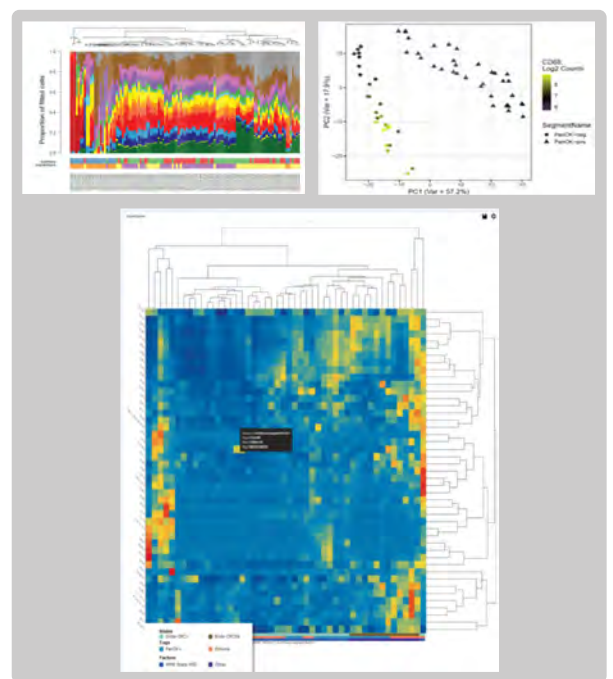
**Image and profile RNA and Protein with GeoMx DSP.**

### 3 Count



Count barcodes on the nCounter® Analysis System or sequence with NGS.

### 4 Analyze

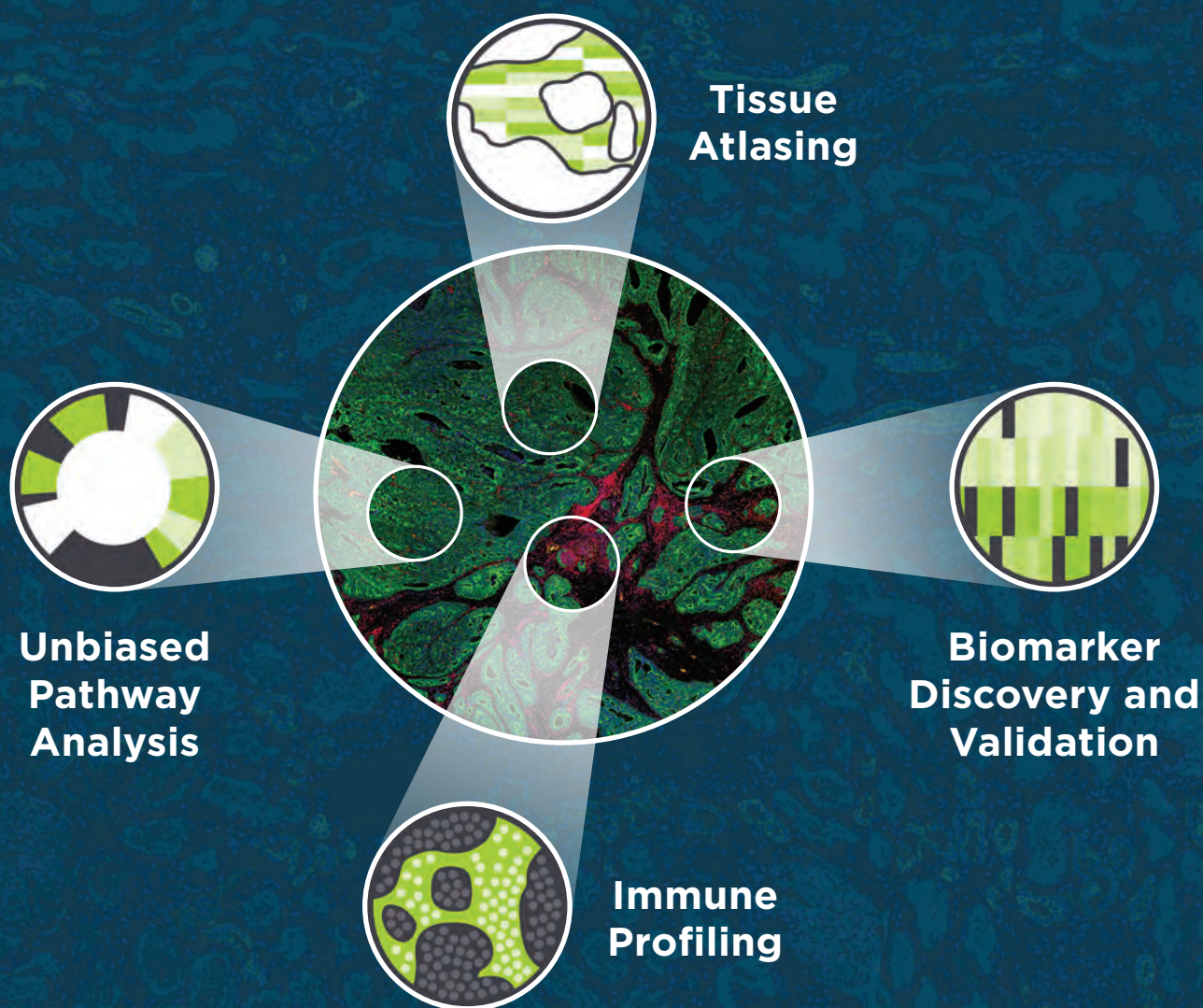


Pre-defined data processing pipelines and interactive data analysis accelerate biological insight.

# Enabling Your Research with the Most Flexible **Spatial Solution**

Whether you are a discovery or translational researcher, the GeoMx<sup>®</sup> DSP is the most flexible spatial solution designed to conform to your ever-changing research needs.

GeoMx DSP combines standard immunofluorescent techniques with molecular profiling technology to perform highly multiplexed, spatially resolved experiments.



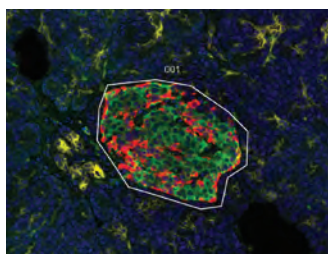
# Locate Your Regions of Interest

## Understand Tissue Structure with Flexibility



### Geometric Profiling

Profile with any geometric shape to characterize distinct tissue regions.

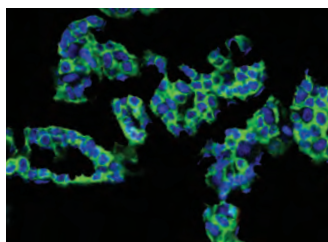


*The Islet of Langerhans is geometrically profiled with Insulin, glucagon and PanCK morphology markers.*



### Segment Profiling

Identify and profile distinct biological compartments within a region of interest (ROI).

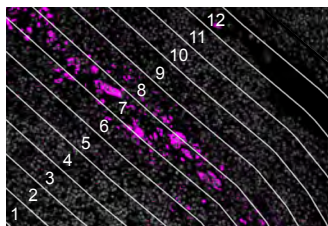


*A 5  $\mu$ m section of kidney depicts the proximal convoluted tubule. This section has been segment profiled guided by CD10 and CD31 morphology marker staining.*

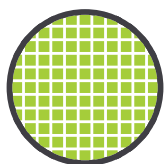


### Contour Profiling

Evaluate how proximity affects biological response and the local microenvironment around a central structure using radiating ROIs.

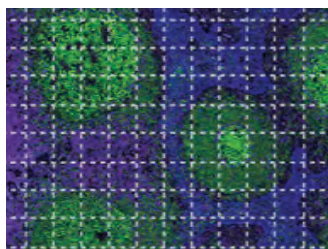


*The invasive margins of two colorectal tumor samples are analyzed using contour segments extending into the tumor and outside the tumor into the stroma and profiled with 1400+ RNA probes with NGS read out.*

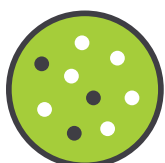


### Gridded Profiling

Perform rigorous spatial mapping using a tunable grid pattern.

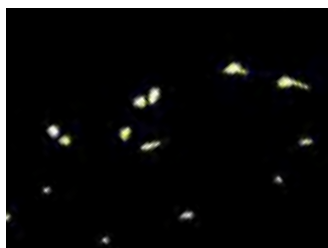


*Gridded protein profiling of a tonsil section, stained with morphology markers CD3, CD20 and PanCK.*



### Cell Type Specific Profiling

Reveal the function of cell populations guided by cell type specific morphology markers.

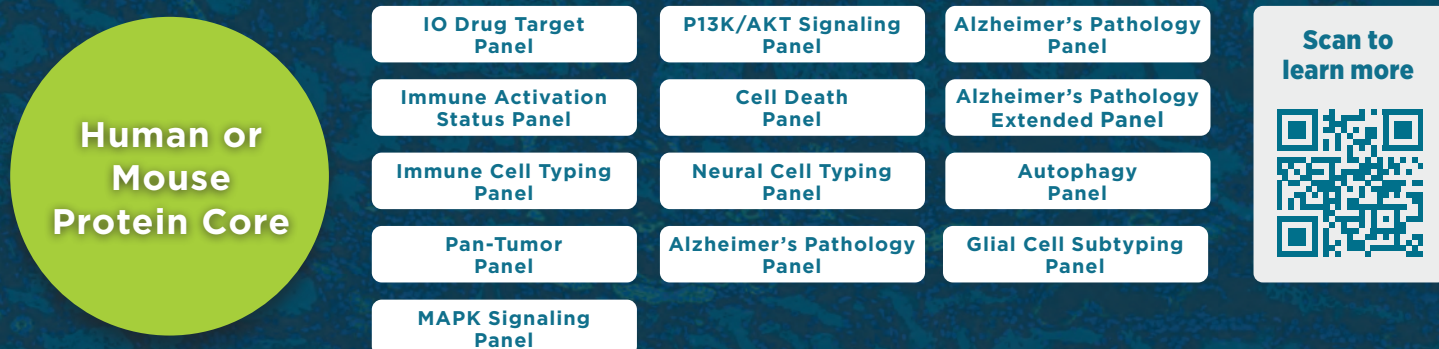


*Paneth Cells from the Colon, stained with morphology marker 5-HT.*

# Flexible, Pre-Validated Protein Content to Fit a Range of Research Needs

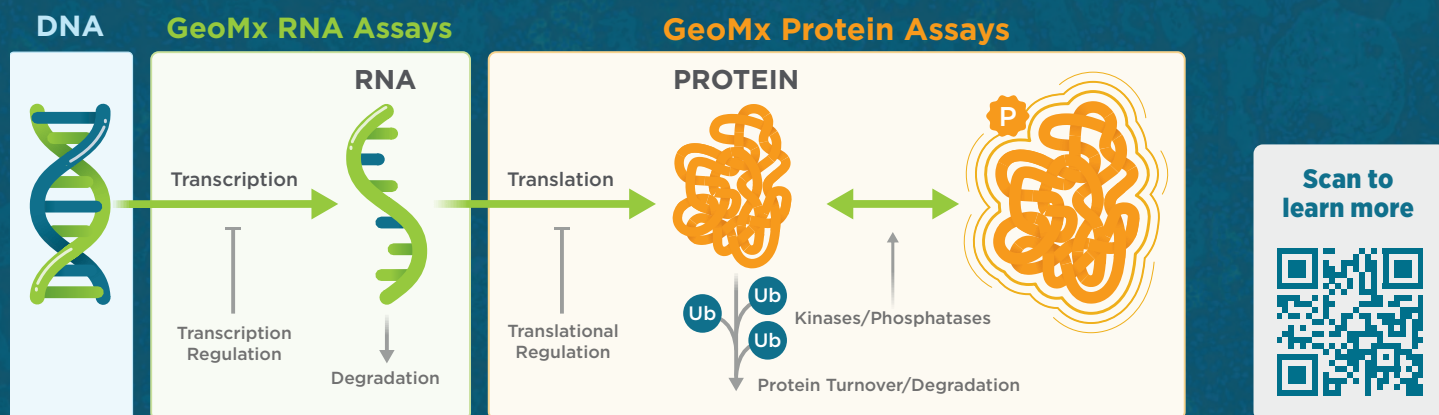
GeoMx® protein assays are modular and optimized for robust performance across a variety of sample types. Choose your readout method and the core of your choice. For readout on the nCounter® Analysis System, add on up to 6 modules plus as many as 10 custom targets of choice. For NGS readout, add on as many different modules as you prefer plus up to 10 custom targets of choice.

**Available content covers immunology, immuno-oncology, and neuroscience with a rapidly growing portfolio.**



## Proteogenomics Meets Spatial Biology with GeoMx DSP

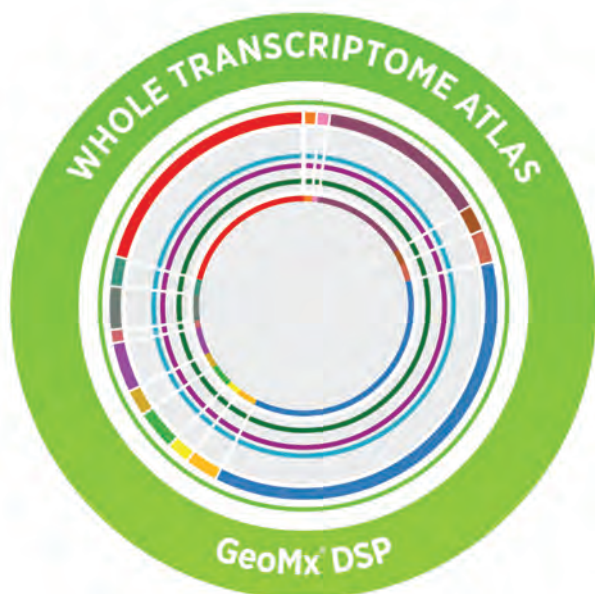
Simplify your spatial proteogenomic workflow with multimodal, co-detection of RNA and Protein from the same tissue section. Combine the GeoMx RNA Assays and GeoMx Protein Assays to gain a complete picture of biology all the way from transcription to protein activation.



# GeoMx<sup>®</sup> Whole Transcriptome Atlases

## Spatial Resolution for Any Target

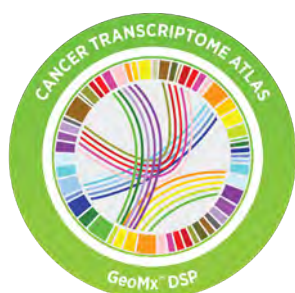
The GeoMx Whole Transcriptome Atlas (WTA) provides an unbiased, spatial view of all protein-coding genes by leveraging the power of NGS. Whether you are mapping the architecture of tissue or exploring the regulation of morphological features, WTA delivers the highest sensitivity for spatial whole transcriptomics on FFPE tissues.



- ✓ **Non-Poly-A pulldown, whole transcriptome assay**
- ✓ **Complete coverage**
- ✓ **Visualize morphology**
- ✓ **Streamline workflows**
- ✓ **Customization**
- ✓ **Minimal sequencing**
- ✓ **Co-detection with protein**

## Explore More GeoMx RNA Assays

GeoMx RNA assays include an immune pathways panel for nCounter readout that allows for profiling of up to 84 human RNAs, including probes for the Tumor Inflammation Signature (TIS) and the ability to add up to 10 targets of interest and 2 additional controls. For higher plex spatial transcriptomics, take advantage of NGS readout and choose from the targeted Cancer Transcriptome Atlas or design a Custom RNA Assay of up to 400 targets from any species.



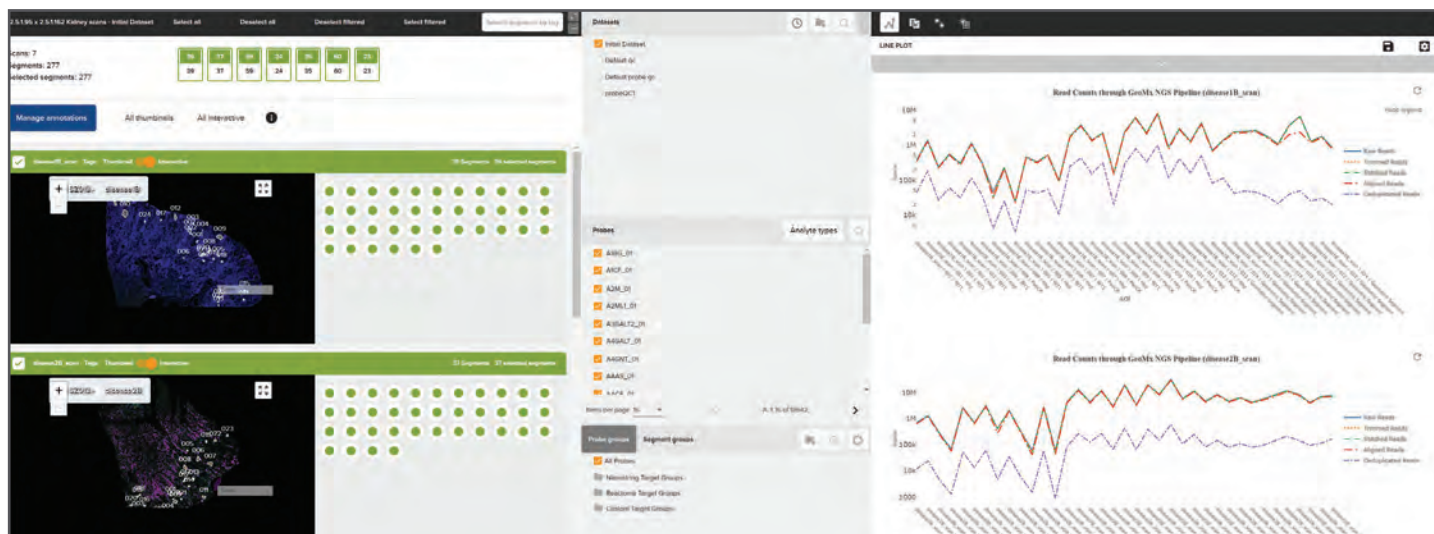
Scan to  
learn more



# Explore Your Data

## GeoMx<sup>®</sup> Data Analysis Suite (DSPDA)

DSPDA is an interactive data analysis suite that connects quantitative data to spatial context to provide a seamless experimental workflow.



**Visualize your counts based on your ROI selection**

**Data QC and normalization**

**Visualize pathway analysis, differential expression, heatmaps and more!**

## GeoScript<sup>™</sup> Hub

NanoString has validated and released code packages to the open-source community; Explore GeoScript Hub to see how these tools can be used to configure data analysis pipelines.

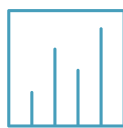
**Normalization for Protein**



**RNA Negative Normalization**



**Spatial Decon**



**Dimension Reduction**



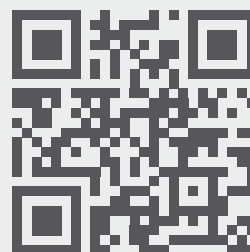
**Volcano Plot**



**Cell-Type Contouring**



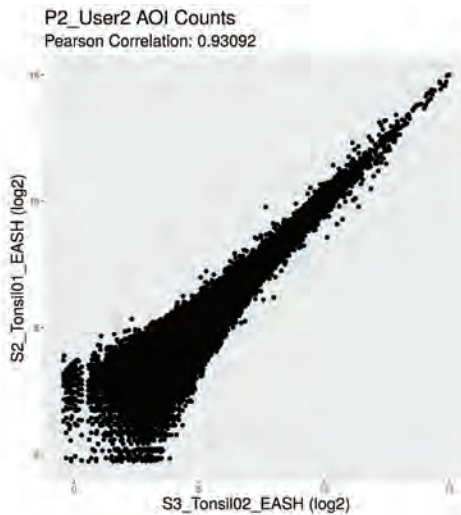
**Scan to learn more about the GeoScript Hub**



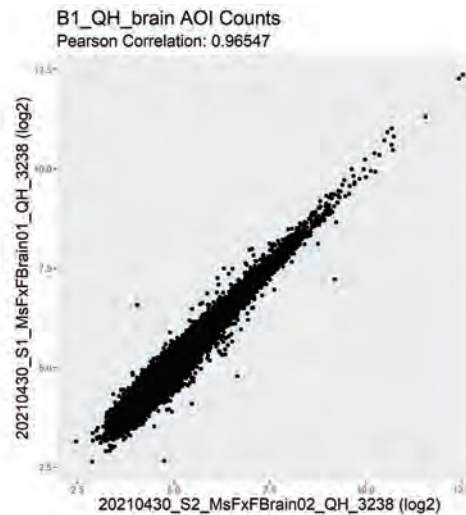
# Consistent Results, Reliable Answers

Multi-Sample Analysis and Cohort Studies Made Easy  
with Unmatched Reproducibility and Scalability

Human FFPE Tonsil



Mouse Fixed Frozen Brain

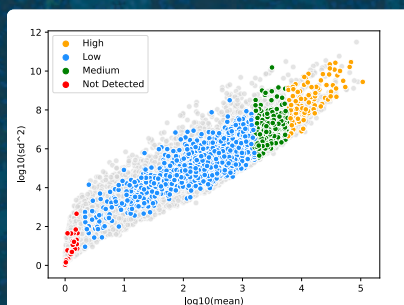


Proven reproducibility  
between slides, users, and  
instruments with the Whole  
Transcriptome Atlas.

## Detect More with Sensitivity

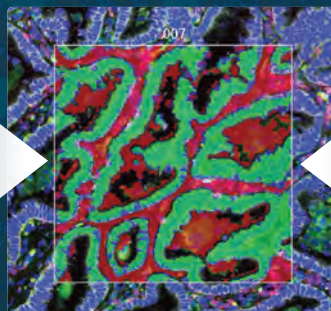
GeoMx® DSP detects the most relevant low to high expressing genes. Detect only the targets that matter  
in the regions that matter.

Dynamic  
Expression Range

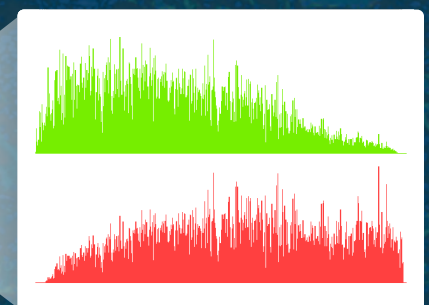


Expression bins defined using  
TCGA NSCLC data

Non-Small Cell Lung Cancer



Tumor Profile



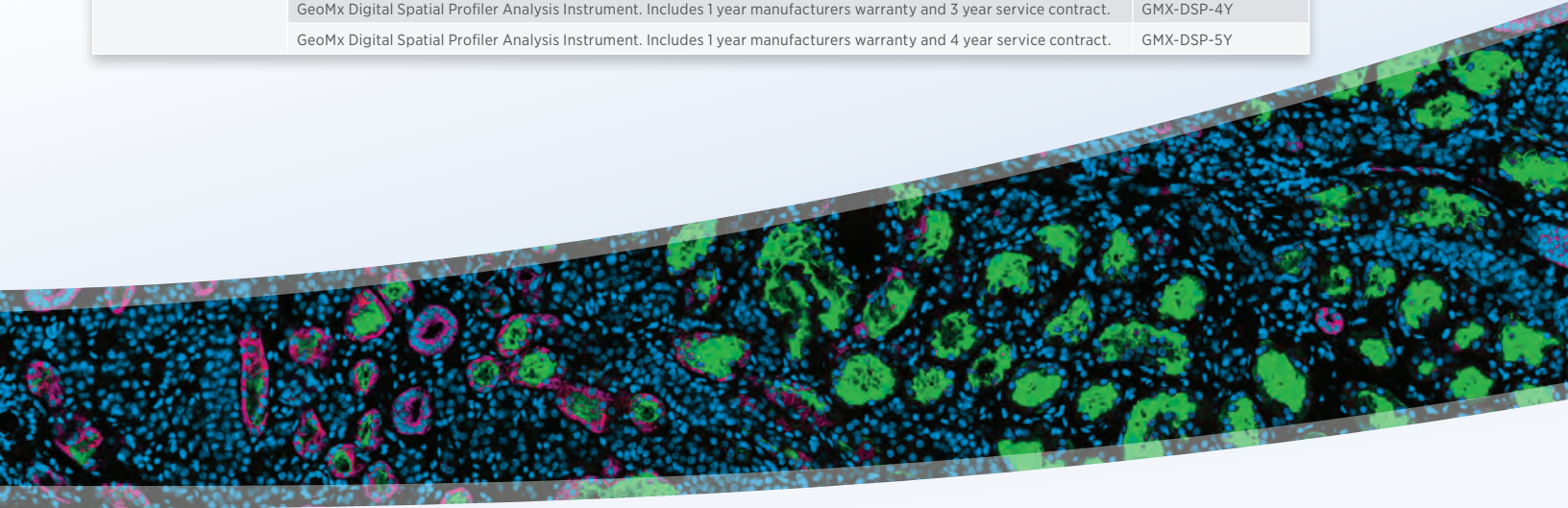
Immune Profile

Specifications

| Category                          | Feature                                          | Specification                                                                                                                                                                                                                                                                         |
|-----------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GeoMx <sup>®</sup> DSP Instrument | Sample Throughput                                | Up to 8 slides/day (at 12 ROIs per slide, 10 mm by 10 mm scan area)                                                                                                                                                                                                                   |
|                                   | Minimum UV Illumination Area                     | 10 µm diameter                                                                                                                                                                                                                                                                        |
|                                   | Resolution                                       | 20X 0.45 NA objective                                                                                                                                                                                                                                                                 |
|                                   | Imaging Modes                                    | Fluorescence                                                                                                                                                                                                                                                                          |
|                                   | Imaging Channels (representative dyes)           | 4 fluorescent channels: (FITC/SYTO13/AF488), (CY3/AF532/PE/SYTO83), (TxRed/AF594), (CY5/AF647/Dylight 650)                                                                                                                                                                            |
|                                   | Imaging channels (em center wavelength/bandpass) | 516/23, 564/15, 623/30, 683/30                                                                                                                                                                                                                                                        |
|                                   | Slide Capacity                                   | Four 1 x 3 inch slides                                                                                                                                                                                                                                                                |
|                                   | On Instrument Data Storage Capacity              | 8TB (> 300 10 mm x 10 mm 4 channel slide images)                                                                                                                                                                                                                                      |
|                                   | Long Term Data Storage                           | Customer-provided fileshare (local network)                                                                                                                                                                                                                                           |
|                                   | ROI Definition                                   | On-instrument or via web browser                                                                                                                                                                                                                                                      |
|                                   | ROI Selection                                    | Geometric, Segmentation, Cell Type Specific Phenotype, Contouring, Gridding                                                                                                                                                                                                           |
|                                   | Instrument Dimensions                            | Actual: 30" x 29" x 24"/76 cm x 73 cm x 61 cm                                                                                                                                                                                                                                         |
|                                   | Instrument Weight                                | 220 lb/100 kg                                                                                                                                                                                                                                                                         |
|                                   | Power source                                     | 110-240 VAC, 50/60Hz, 440VA                                                                                                                                                                                                                                                           |
|                                   | Readout Instrument Compatability                 | nCounter Analysis System, Illumina NGS                                                                                                                                                                                                                                                |
|                                   | Image Export                                     | Single-channel pyramidal TIFF; monochrome or color images (JPEG, PNG, WEBP); multichannel pyramidal, stitched OME- TIFF                                                                                                                                                               |
| GeoMx DSP Reagents                | Supported Analytes                               | Protein and RNA                                                                                                                                                                                                                                                                       |
|                                   | Chemistry Multiplexing Platform Capabilities     | Up to 96 plex for nCounter, 20K plex for NGS                                                                                                                                                                                                                                          |
| GeoMx Data Analysis Software      | Data Visualization and Analysis                  | Intuitive and interactive interface that automatically connects quantitative readout with spatial information. Workflow includes QC and normalization. Visualization include clusters, heatmaps, volcano plots, bar graphs, box plots, strip plots, scatter plots, correlation plots. |
|                                   | Data Export                                      | .xlsx file format for raw or calibrated data                                                                                                                                                                                                                                          |
|                                   | Image Export                                     | .svg format for visualization plots                                                                                                                                                                                                                                                   |

Instrument Information

| Product                        | Description                                                                                                             | Catalog Number |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------|
| GeoMx Digital Spatial Profiler | GeoMx Digital Spatial Profiler Analysis Instrument. Includes 1 year manufacturers warranty.                             | GMX-DSP-1Y     |
|                                | GeoMx Digital Spatial Profiler Analysis Instrument. Includes 1 year manufacturers warranty and 1 year service contract. | GMX-DSP-2Y     |
|                                | GeoMx Digital Spatial Profiler Analysis Instrument. Includes 1 year manufacturers warranty and 2 year service contract. | GMX-DSP-3Y     |
|                                | GeoMx Digital Spatial Profiler Analysis Instrument. Includes 1 year manufacturers warranty and 3 year service contract. | GMX-DSP-4Y     |
|                                | GeoMx Digital Spatial Profiler Analysis Instrument. Includes 1 year manufacturers warranty and 4 year service contract. | GMX-DSP-5Y     |



For more information, visit [nanosttring.com/GeoMxDSP](https://nanosttring.com/GeoMxDSP)

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